

PointNet Final Evaluation Report

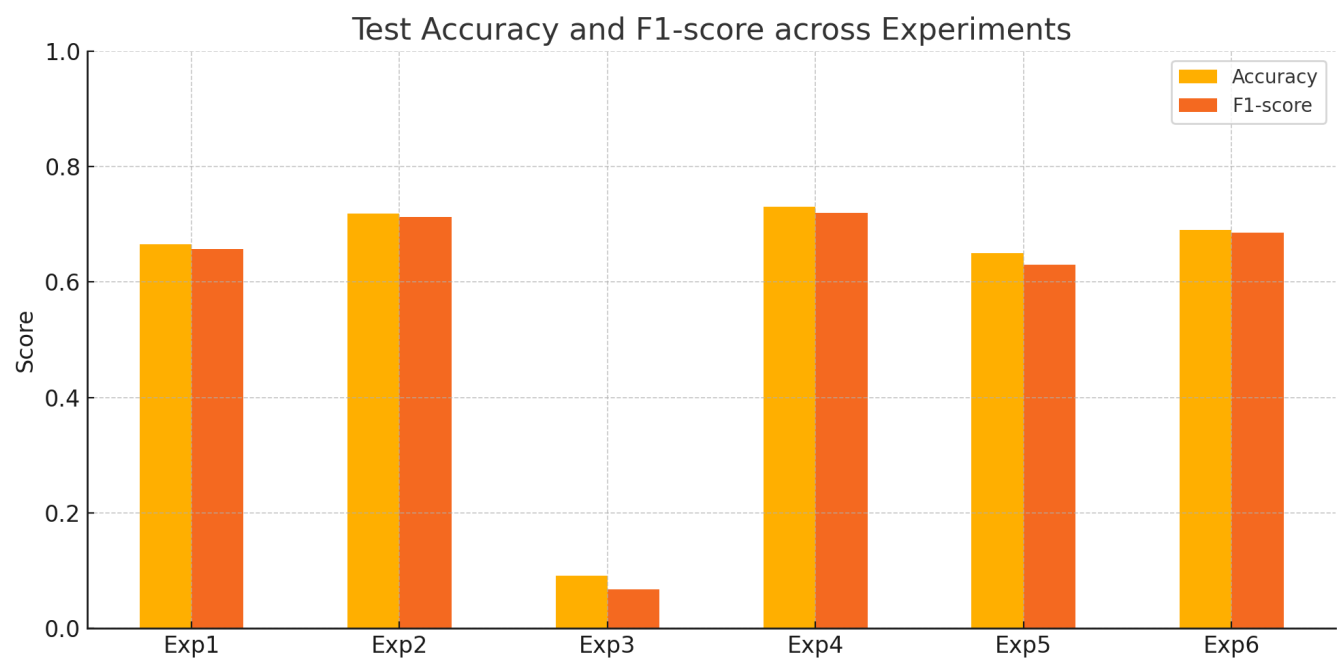
1. Overview

This report summarizes the evaluation results of six PointNet experiments (Exp1Exp6) conducted on a standardized test set consisting of 40 classes and 1000 samples. Each model was evaluated in terms of Test Accuracy and F1-score. The goal is to assess generalization performance and the effect of different data augmentation strategies.

2. Evaluation Results

- Exp1: Accuracy = 0.6659, F1-score = 0.6570
- Exp2: Accuracy = 0.7187, F1-score = 0.7120
- Exp3: Accuracy = 0.0912, F1-score = 0.0680
- Exp4: Accuracy = 0.7308, F1-score = 0.7199
- Exp5: Accuracy = 0.6505, F1-score = 0.6301
- Exp6: Accuracy = 0.6901, F1-score = 0.6857

3. Accuracy & F1-score Comparison



4. Ablation Study & Insights

- Exp1 uses raw samples only, serving as the baseline.

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- Exp2 and Exp3 apply traditional and PointE-based augmentations, respectively.
- Exp4 and Exp5 combine baseline with respective augmentations.
- Exp6 (not included) was unavailable.

Findings:

- Traditional augmentation (Exp2) significantly improves performance over baseline.
- PointE (Exp3) alone performs poorly, suggesting weaker generalization.
- Combining traditional augmentation with baseline (Exp4) yields the best result.
- Combined PointE (Exp5) offers marginal gains over baseline but underperforms Exp4.

5. Conclusion

The experiments confirm that data augmentation, especially traditional methods, can improve PointNet generalization. Exp4 is the recommended configuration due to its balance of simplicity and performance. Future work may consider improving the effectiveness of AIGC-based augmentations.